OBJECT ORIENTED PROGRAMMING LAB
ASSIGNMENT-4
(Topic – Inheritance)

Programming Questions:

1. **Create a C++ program to demonstrate Single Inheritance.**
   Create a base class Person with the following data members:
   Name
   Age
   Create a derived class Student having:
   Roll Number
   Course
   Implement member functions to accept and display the complete details of a student using inheritance.

2. **Create a C++ program to demonstrate Multilevel Inheritance.**
   Create three classes:
   Person → stores Name
   Employee → stores Employee ID
   Manager → stores Department
   Derive Employee from Person and Manager from Employee. Display all the details of the manager.

3. **Create a C++ program to demonstrate Multiple Inheritance.**
   Create two base classes:
   Academic → stores Marks
   Sports → stores Sports Score
   Create a derived class Result that inherits from both classes and calculates the total score and displays the result.

4. **Create a C++ program to demonstrate Hierarchical Inheritance.**
   Create a base class Vehicle containing:
   Brand
   Model
   Create two derived classes:
   Car → Number of Doors
   Bike → Engine Capacity
   Create separate objects of Car and Bike and display their respective details.

5. Create a **base class College** with a constructor that initializes the College Name. Create a **derived class Department** with a constructor that initializes the Department Name.

   Create an object of the derived class and display both college and department information to show the order of constructor execution.